PARISHRAM 2.0 2026 (Class 12th)

Biology (Botany)

DPP: 01

SEXUAL REPRODUCTION IN FLOWERING PLANTS

Q1 What is the process by which the cells of the sporogenous tissue give rise to microspore tetrads during anther development?

(A) Mitosis (B) Meiosis

(C) Binary fission (D) Budding

Q2 Where is the proximal end of the filament attached?

P) To the ovary

Q) To the petal or thalamus

R) To the stigma

S) To the sepal

(A) P only (B) P and Q

(C) Q only (D) P, R and S

Q3 In a typical angiosperm anther, how many microsporangia are present, and where are they located?

(A) Two, one in each lobe

(B) Four, two in each lobe

(C) Three, one in each theca

(D) Six, three in each lobe

Q4 Which of the following statements is correct ?

(A) A typical angiospermic anther is dithecous and tetralobed.

(B) The bilobed nature of an anther is very distinct in the longitudinal section of the anther.

(C) The anther is a tetragonal structure consisting of two microsporangia.

(D) The microsporangia mature and transform into pollen sacs.

Q5 Which of the following correctly represents the wall layers of a typical microsporangium from outermost to innermost?

(A) Epidermis Middle layers Endothecium Tapetum

(B) Epidermis Endothecium Middle layers Tapetum

(C) Tapetum Endothecium Middle layers Epidermis

(D) Middle layers Epidermis Endothecium Tapetum

Q6 Statement I**:** The tapetum is the innermost wall layer of the microsporangium and plays a crucial role in nourishing developing pollen grains.

Statement II: Tapetal cells possess dense cytoplasm and often have more than one nucleus, which enhances their ability to support pollen development.

(A) Both statements are correct

(B) Both statements are incorrect

(C) Statement I is correct and statement II is incorrect

(D) Statement I is incorrect and statement II is correct

Q7 **Assertion:** Pollen grains are well-preserved as fossils.

Reason: The outer layer of the pollen grain, called the exine, is made of sporopollenin, which is highly resistant to extreme conditions and degradation.

(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(C) Assertion is true, but Reason is false.

(D) Assertion is false, but Reason is true.

Q8 Which of the following statements correctly explains the factors affecting pollen viability and its preservation?

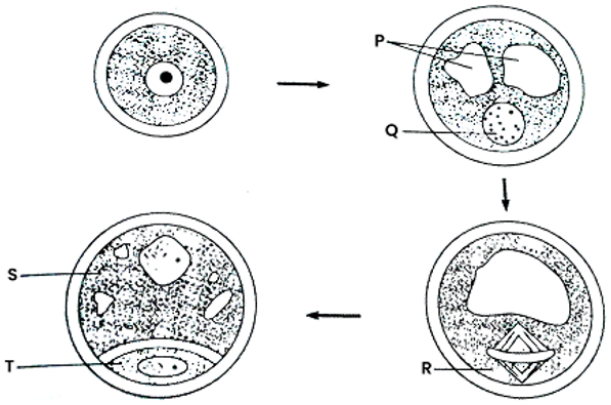
(A) Pollen grains retain viability for a fixed period, regardless of environmental conditions.

(B) In cereals like rice and wheat, pollen viability lasts for several months due to their adaptation to extreme climates.

(C) Pollen grains of certain plant families, such as Rosaceae and Solanaceae, can remain viable for months, while cereals like rice and wheat lose viability within minutes.

(D) Pollen grains cannot be stored for long periods, even under extremely low temperatures like liquid nitrogen (-196°C).

Q9 Choose the correct option for the given diagram



(A) T is an asymmetric spindle.

(B) P and Q are vacuoles.

(C) S is a generative cell

(D) S is vegetative cell

Q10 Which of the following statements is correct regarding pollen development and its effects?

(A) In all angiosperms, pollen grains are shed at the 3-celled stage.

(B) Over 60% of angiosperms release pollen at the 3-celled stage, while the rest release it at the 2-celled stage.

(C) Over 60% of angiosperms release pollen at the 2-celled stage, while the rest release it at the 3-celled stage.

(D) Parthenium (carrot grass) is beneficial for human health and does not contribute to pollen allergies.

Answer Key

Q1	B
Q2	C
Q3	B
Q4	D
Q5	B
Q6	A
Q7	A
Q8	C
Q9	D
Q10	C

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 **Text Solution:**

Meiosis

Video Solution:

Q2 **Text Solution:**

To the petal or thalamus

Video Solution:

Q3 **Text Solution:**

Four, two in each lobe

Video Solution:

Q4 **Text Solution:**

The microsporangia mature and transform into pollen sacs.

Video Solution:

Q5 **Text Solution:**

Epidermis Endothecium Middle layers Tapetum

Video Solution:

Q6 **Text Solution:**

Both statements are correct

Video Solution:

Q7 **Text Solution:**

Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Video Solution:

Q8 **Text Solution:**

Pollen grains of certain plant families, such as Rosaceae and Solanaceae, can remain viable for months, while cereals like rice and wheat lose viability within minutes.

Video Solution:

Q9 **Text Solution:**

S is vegetative cell

Video Solution:

Q10 **Text Solution:**

Over 60% of angiosperms release pollen at the 2-celled stage,
while the rest release it at the 3-celled stage.

Video Solution:

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